João Ricardo Silva Matos

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Lisbon, Portugal

Education

- **PhD. Student in Computer Science** — *September 2026 - August 2031*
  Instituto Superior Técnico, University of Lisbon
  School of Computer Science, Carnegie Mellon University
  Lisbon, Portugal; Pittsburgh, USA
  - Dual degree program jointly conferred by IST and CMU
  - Member of the Automated Reasoning and Software Reliability (ARSR) group at INESC-ID

- **MSc. in Computer Science and Engineering** — *September 2023 - November 2025*
  Instituto Superior Técnico, University of Lisbon
  Lisbon, Portugal
  - Double specialization in Algorithms and Cybersecurity
  - Worked under the supervision of Prof. Pedro Adão
  - Dissertation: *"Controlling Path Explosion in x86 Symbolic Execution"* (19/20) [🌐]
  - Final grade: 18/20

- **Erasmus+** — *February - July 2024*
  Akademia Górniczo-Hutnicza (AGH University of Kraków)
  Kraków, Poland
  - Exchange semester under the Erasmus+ program
  - Completed certification for A1 level in Polish

- **BEng. in Informatics Engineering** — *September 2020 - July 2023*
  Faculty of Sciences, University of Lisbon
  Lisbon, Portugal
  - Took two extra courses on programming competitions
  - Final grade: 17/20

Other Academic Activities

- **BIP Unite short course in Cybersecurity** — *October - November 2025*
  Université Grenoble Alpes
  Grenoble, France
  - Participated with peers on a case study about cache template attacks
  - Worked under the supervision of Prof. Daniel Gruß
  - Grade: A

- **BIP Unite short course in Artificial Intelligence** — *September - October 2024*
  Université Grenoble Alpes
  Grenoble, France
  - Worked on a case study on sustainable AI
  - Grade: 19/20

Teaching Experience

- **Instituto Superior Técnico [🌐]** — *Various semesters, Sept. 2024 - Now*
  Teaching Assistant
  Lisbon, Portugal
  - Forensic Cybersecurity - MSc (2026/27; 2025/26; 2024/25):
    - Taught weekly lab classes
    - Prepared and graded the course project
  - Software Security - MSc (2026/27; 2025/26):
    - Taught two lab classes per week
    - Graded the course project and respective written test
  - Introduction to Computer Architecture - BSc (2026/27):
    - Will teach lab classes in the second semester

Experience

- **ThinkOpen Solutions [🌐]** *Sept. 2025 - Aug. 2026*
  Software Developer Oeiras, Portugal/Remote
  - Worked as a software developer for the Odoo platform
  - The development team worked on custom modules according to customer needs
  - Projects involved Python, XML, and JavaScript
  - Also performed system administration duties

- **Premium Minds [🌐]** *July - August 2023*
  Software Engineering Intern Lisbon, Portugal
  - Developed an internal application for employee timekeeping and salary calculations
  - Application was created using Java and NextJS

Projects

- **Externalization of concretization and state merging policies in symbolic execution** *Aug. 2024 - Nov. 2025*
  Tool: AVD symbolic executor
  - Implemented a modified version of the CS_{ML} language for concretization policies
  - Created a new language based on CS_{ML} for the specification of merge policies, MC_{ML}
  - Modified AVD, a symbolic engine created at Técnico, adding support for these languages
  - Created a VSCode extension that supports linting and syntax highlighting for these languages [🐙][🌐]
  - Created a testing framework for symbolic execution engines

Honors and Awards

- **Instituto Superior Técnico Academic Excellence Award** *2026*
  Instituto Superior Técnico, University of Lisbon [🌐]
  - Awarded for my academic performance in the 2024/25 academic year
  - Award is given every academic year to the top 10% of students (ECTS grade A) with a yearly average of at least 18/20

- **Instituto Superior Técnico Academic Merit Award** *2025*
  Instituto Superior Técnico, University of Lisbon [🌐]
  - Awarded for my academic performance in the 2023/24 academic year
  - Award is given every academic year to the top 25% of students (ECTS grade B) with a yearly average of at least 17/20

- **Faculty of Sciences Merit Award** *2022; 23*
  Faculty of Sciences, University of Lisbon [🌐]
  - Awarded twice for my academic performance in the 2020/21 and 2021/22 academic years
  - Award is given to students with a year average higher than 16/20 (1st year) or 16.5/20 (2nd year)

Other Activities

- **Participated in the Guinness World Records' world's largest programming class** *2024*
  Instituto Superior Técnico, University of Lisbon
- **Member of the Ribeiro Santos student dorm board** *2022-2023*
  University of Lisbon
- **Competed in the *MIUP'21* national university programming competition** *2021*
  Instituto Superior Técnico, University of Lisbon

Additional Information

Languages: Portuguese (native), English (fluent, certified C1), Spanish (fluent), Polish (basic, certified A1)
Interests: Computability; Computational Complexity; Cybersecurity; Formal Verification

REFERENCES

1. **Pedro Adão**
   Associate Professor, Department of Computer Science and Engineering
   Instituto Superior Técnico
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   MSc. & PhD. Thesis Advisor

2. **José Fragoso**
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   MSc. Thesis Reviewer & PhD. Co-Advisor

3. **Ruben Martins**
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4. **Nuno Santos**
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   Forensic Cybersecurity Lecturer